



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

CHARCOAL ADSORBER BLOWDOWN KNOCKOUT DRUM (D-2211)

Docu. No. : 14780-8120-PS-D2211
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
Page : 1 / 4

TITLE :

CHARCOAL ADSORBER BLOWDOWN KNOCKOUT DRUM

(D-2211)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-D2211
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 22, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	DEC 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 2, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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VESSELS

PROJECT SPECIFICATION

963765 — 301

SHEET 3 / 4

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	27APR12	TS	HDT				

General Notes

- A. Vessels shall be constructed in accordance with the following unless otherwise specified on the individual vessel specification sheet.

Standards:

UOP Standard Specification 3-11

- B. Although the resulting inside diameter may be slightly different than specified, vessels without trays or packing, may be fabricated from pipe.
- C. The dimensioned location of some of the vessel level nozzles may be noted as min (minimum) or max (maximum). It is the responsibility of the contractor to adjust these dimensions to suit the specific level requirements specified on the UOP Piping and Instrument Diagram(s) and on UOP Standard Drawing 8-121. When additional nozzles are required for level instrumentation, the contractor shall specify them consistent with the size, class and facing requirements specified in the UOP Project Specification 301-Vessels and on the UOP Piping and Instrument Diagram(s).
- D. When system of measurement is metric, all dimensions are in millimeters unless noted otherwise.
- E. Contractor shall not locate TI thermowells in a downcomer. Contractor shall also ensure that TI thermowells are located to avoid interference with downcomers, weirs, distributors, collectors, baffles, chimneys, traps and all other vessel internals.
- F. Suppliers (or equal):

Spray Nozzle:

BETE Fog Nozzle Inc.
www.bete.com

Profile Wire Screens:

Johnson Screens
www.johnsonscreens.com

Nagaoka International Corporation
www.nagaokajapan.co.jp

Other specifications
referenced in this
specification:

UOP Standard Specification: 3-11, 3-15, 3-35, 9-11

UOP Standard Drawing: 3-115, 3-116, 3-122, 3-123, 3-144, 3-183, 3-190, 3-264, 3-266, 3-313, 3-322, 8-121

UOP Project Specification: 963765-106, 963765-307



PROJECT SPECIFICATION

963765 - 301

SHEET 4 / 4

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VESSELS

REV	DATE	BY	APP'D	REV	DATE	BY	APP'D
0	07SEP12	JG	HDT				

SERVICE CHARCOAL ADSORBER COOLDOWN KNOCKOUT DRUM

ITEM NO D-2211

Design	INT	3.5 kg/cm2(g)	@ 120°C
Conditions	EXT	FULL VACUUM	@ 250°C
at Top	Metal Temperature(Min)		@ -10°C
Operating		0.057 kg/cm2(g)	@ 5°C
Conditions			@
Radiograph MR			
Postweld Heat Treat MR			

	Material	Corrosion Allowance
	SA 240 TYPE 304L	1.5

This Vessel is in the Following Service(s):

Hydrogen	NO
Hydrofluoric (HF) Acid	NO
Hydrofluoric (HF) Acid (with Resid. Elem. Control)	NO
Wet Hydrogen Sulfide (H2S)	NO
Cyclic	NO
Caustic	NO
Carbonate	NO
Amine	NO

Vessel Heads	
Top	2:1 ELLIPTICAL
Bottom	2:1 ELLIPTICAL

Accessories Applied by Fabricator	
Ladders & Platform Clips	AS REQUIRED
Insulation Clips and Rings	C (50) YES
Vessel Support	LEGS YES

Nozzles and Manways			
Mark	No	Size inches	Service
G01	1	4	VENTILATION
O01	1	36	OUTLET
V01	1	2	VENT
B02	1	24 ID	MANWAY
J02	1	44	INLET
L02	2	2	LG (2)
L03	2	2	LT (2)
S02	1	2	STEAMOUT (2)
W03	1	3	OUTLET

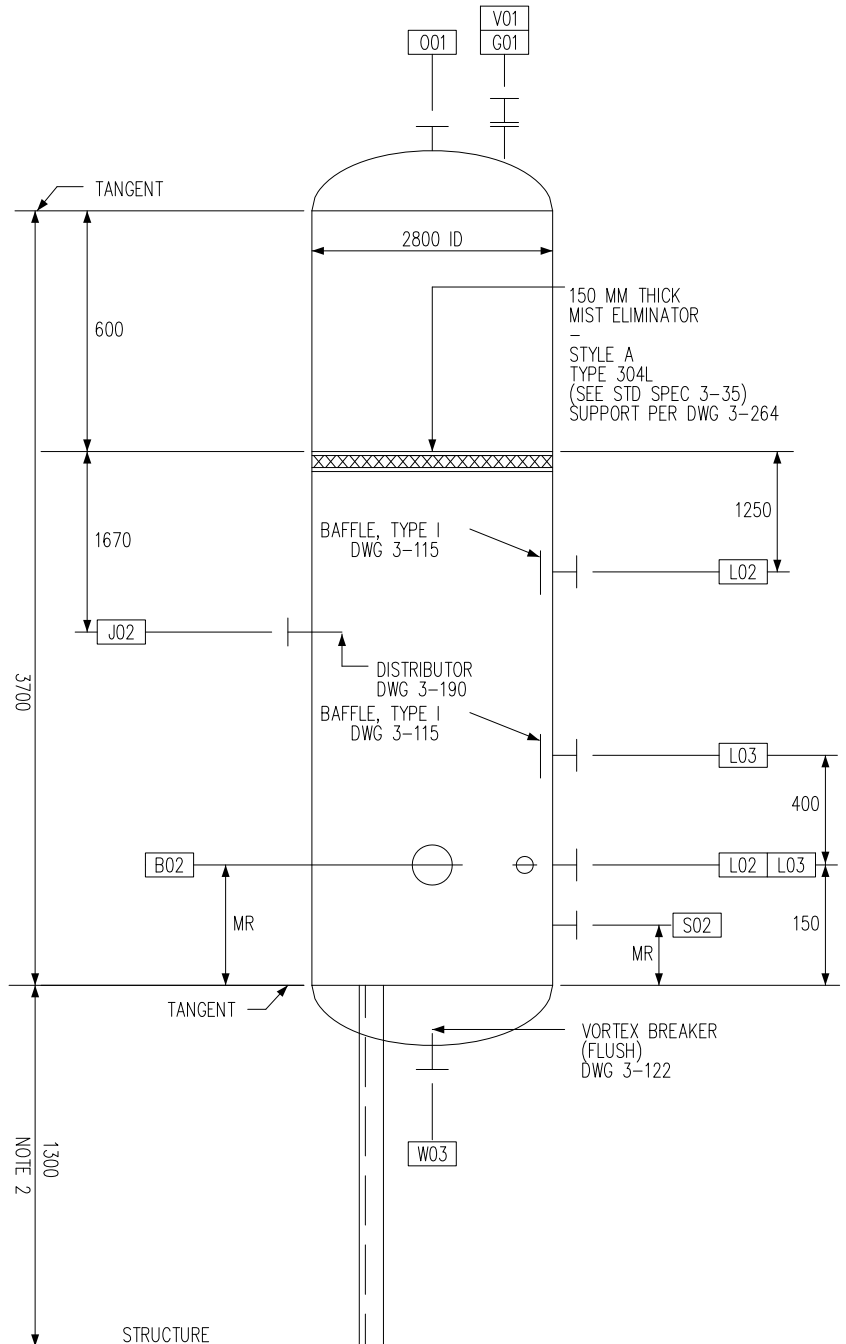
Class - ASME CL150

(2) ASME CL300

Facing - RAISED FACE

Normal Liquid Level = 350 mm ABOVE BOTTOM TANGENT

Specific Gravity = 0.9064



NOTES:

- 1) VESSEL SHALL BE CONSTRUCTED IN ACCORDANCE WITH UOP STD. SPEC 3-15. ALTERNATELY: THE ENTIRE VESSEL MAY BE KILLED CARBON STEEL, AND LINED WITH 3mm THICK TYPE 304L. ALL INTERNALS TO BE STAINLESS STEEL. ALL NOZZLES AND MANWAYS TO BE LINED AND FACED WITH STAINLESS STEEL. LINING IS CORROSION ALLOWANCE. KILLED CARBON STEEL VESSEL SHALL BE CONSTRUCTED IN ACCORDANCE WITH UOP STANDARD SPECIFICATION 3-11.
- 2) VESSEL ELEVATION MUST BE 2500 mm MINIMUM ABOVE TOP OF DECANter (ITEM D-2205).

Drawings Referred to in this Specification

3-115	3-122	3-190	3-264			
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